A B C A L C U L U S

CHAPTER 6

Integration & Accumulation of Change

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **6.1 – 6.10, 6.14** AB

+ Extensions: **6.11** IBP · **6.12** Partial Fractions · **6.13** Improper Integrals BC

Name: _____ Period: _____

PART I

Topics 6.1 – 6.10

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 6.2 Riemann Sums No Calculator

The interval $[0, 4]$ is divided into four subintervals of equal length. Which expression gives the right Riemann sum approximation for $\int_0^4 f(x)\,dx$?

(A) $f(0) + f(1) + f(2) + f(3)$

(B) $f(1) + f(2) + f(3) + f(4)$

(C) $f(0.5) + f(1.5) + f(2.5) + f(3.5)$

(D) $\frac{1}{2}\left[f(0) + 2f(1) + 2f(2) + 2f(3) + f(4)\right]$

Q2 EASY 6.6 Properties No Calculator

If $\int_0^5 f(x)\,dx = 12$ and $\int_0^2 f(x)\,dx = 3$, then $\int_2^5 f(x)\,dx =$

(A) -9

(B) 9

(C) 15

(D) 36

Q3 EASY 6.4 FTC No Calculator

If $F(x) = \int_1^x t^2\,dt$, then $F'(x) =$

(A) $\frac{x^3}{3}$

(B) $x^2 - 1$

(C) x^2

(D) $2x$

Q4 MEDIUM 6.4 FTC + Chain No Calculator

If $F(x) = \int_0^{x^2} \sin t\,dt$, then $F'(x) =$

(A) $\sin(x^2)$

(B) $2x\sin(x^2)$

(C) $-\cos(x^2) + 1$

(D) $\cos(x^2) \cdot 2x$

Q5 EASY 6.8 Antiderivatives No Calculator

$\int (3x^2 - 4x + 5)\,dx =$

(A) $x^3 - 2x^2 + 5x + C$

(B) $x^3 - 2x^2 + 5 + C$

(C) $6x - 4 + C$

(D) $\frac{x^3}{3} - \frac{4x^2}{2} + 5x + C$

Q6 MEDIUM 6.9 U-Substitution No Calculator

$$\int 2x\,(x^2+1)^4\,dx =$$

(A) $\dfrac{(x^2+1)^5}{5} + C$

(B) $\dfrac{(x^2+1)^5}{10} + C$

(C) $2x \cdot \dfrac{(x^2+1)^5}{5} + C$

(D) $\dfrac{x^2(x^2+1)^5}{5} + C$

Q7 MEDIUM 6.7 FTC Evaluation No Calculator

$$\int_0^{\pi/2} \cos x \, dx =$$

(A) 0

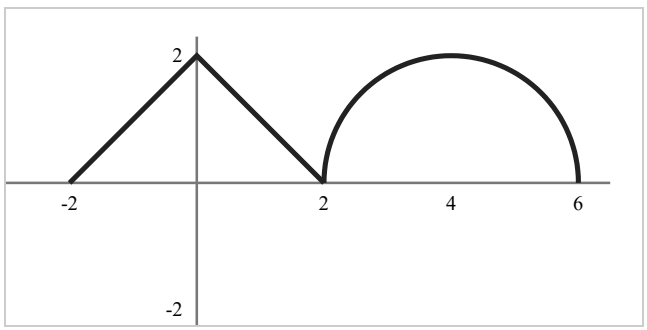
(B) $\frac{1}{2}$

(C) 1

(D) $\frac{\pi}{2}$

Q8 MEDIUM 6.6 Geometry of Integral No Calculator

The graph of f consists of two line segments (forming a triangle above the x-axis with vertices $(-2, 0)$, $(0, 2)$, $(2, 0)$) and a semicircle of radius 2 below the x-axis on $[2, 6]$. Find $\int_{-2}^{6} f(x) \, dx$.



(A) $4 - 2\pi$

(B) $4 + 2\pi$

(C) $3 + 2\pi$

(D) $3 - 2\pi$

Q9 HARD 6.9 U-Sub Definite No Calculator

$$\int_0^1 \frac{x}{(x^2+1)^2} \, dx =$$

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{1}{8}$

(D) $\frac{1}{6}$

Q10 MEDIUM 6.3 Riemann ↔ Integral No Calculator

Which definite integral equals $\lim_{n\to\infty}\sum_{i=1}^{n}\left(1+\frac{2i}{n}\right)^2\cdot\frac{2}{n}$?

(A) $\int_0^2 x^2\,dx$

(B) $\int_1^3 x^2\,dx$

(C) $\int_1^2 (1+x)^2\,dx$

(D) $\int_0^1 (1+2x)^2\,dx$

Q11 HARD 6.2 Over/Underestimate No Calculator

f is positive, increasing, and concave down on $[a,b]$. Which of the following must be an underestimate of $\int_a^b f(x)\,dx$?

(A) Left Riemann sum

(B) Right Riemann sum

(C) Trapezoidal sum

(D) Midpoint Riemann sum

Q12 MEDIUM 6.10 Long Division No Calculator

$$\int \frac{x^2+1}{x+1}\,dx =$$

(A) $\frac{x^2}{2} - x + 2\ln|x+1| + C$

(B) $x - \ln|x+1| + C$

(C) $\frac{x^3+x}{x^2+x} + C$

(D) $\frac{x^2}{2} + \ln|x+1| + C$

Q13 HARD 6.2 / 6.4 Accumulation No Calculator

Selected values of the differentiable function f are given. Let $g(x) = \int_0^x f(t)\,dt$.

x	0	2	4	6	8
$f(x)$	5	3	-1	-4	-2

Using a midpoint Riemann sum with two subintervals of equal length, the approximation for $g(8)$ is

(A) -4

(B) -2

(C) 4

(D) 12

Q14 MEDIUM 6.5 Behavior of g No Calculator

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and is positive on $(0, 3)$, zero at $x = 3$, and negative on $(3, 5)$. At what value of x does g attain its maximum on $[0, 5]$?

(A) 0

(B) 3

(C) 5

(D) Cannot be determined

Q15 EASY 6.1 Accumulation Units No Calculator

Water flows into a tank at a rate $r(t)$ gallons per minute, where t is in minutes. The most appropriate units of $\int_0^{10} r(t)\,dt$ are

(A) gallons per minute

(B) gallons per minute squared

(C) gallons

(D) minutes per gallon

Q16 HARD 6.10 Completing the Square No Calculator

$$\int \frac{1}{x^2 + 2x + 5}\,dx =$$

(A) $\frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C$

(B) $\arctan(x+1) + C$

(C) $\ln|x^2 + 2x + 5| + C$

(D) $\frac{1}{4}\ln|x^2 + 2x + 5| + C$

Q17 MEDIUM 6.14 Selecting a Technique No Calculator

Which technique is most appropriate for evaluating $\int x \sec^2(x^2)\, dx$?

(A) Power rule (direct antidifferentiation)

(B) u -substitution with $u = x^2$

(C) Long division

(D) Completing the square

Q18 MEDIUM 6.6 Properties No Calculator

If $\int_1^4 \left[\, 2f(x) - 3 \,\right] dx = 14$ and $\int_1^4 f(x)\, dx = ?$

(A) $\dfrac{17}{2}$

(B) $\dfrac{23}{2}$

(C) $\dfrac{11}{2}$

(D) 7

PART II *Show All Work*

Free-Response Questions

Free-response answers must include all setup. When using a calculator, present the integral expression with limits and a differential before evaluating. When without a calculator, show antiderivatives, the constant of integration on indefinite integrals, and careful use of substitution.

FRQ 1 EASY 6.7 / 6.8 / 6.9 Evaluation No Calculator

Evaluate each of the following. Show all work.

(a) $\displaystyle\int_1^2 (4x^3 - 6x + 1)\,dx$

(b) $\displaystyle\int \cos(3x)\,dx$

(c) $\displaystyle\int x\,e^{x^2}\,dx$

FRQ 2 MEDIUM 6.1 / 6.2 / 6.3 Reservoir Calculator

The rate at which water enters a reservoir is modeled by the differentiable function $R(t)$, where R is in thousands of gallons per hour and t is in hours since midnight. Selected values of R:

t (hr)	0	3	6	9	12
$R(t)$	5.2	6.8	8.1	7.4	4.5

(a) Use a left Riemann sum with the four subintervals shown to approximate $\int_0^{12} R(t)\,dt$. Using correct units, explain the meaning of this integral in context.

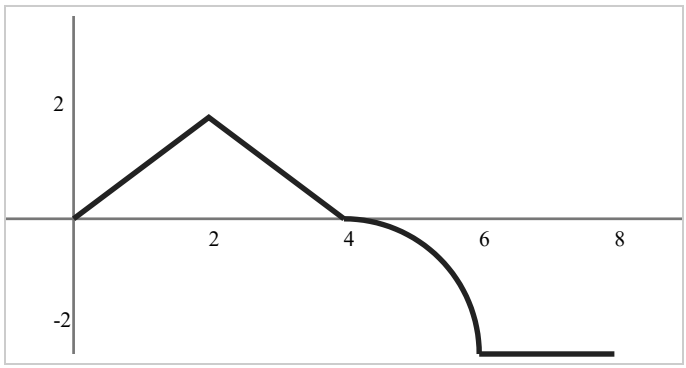
(b) Use a trapezoidal sum with the four subintervals shown to approximate $\int_0^{12} R(t)\,dt$.

(c) Is the left Riemann sum in part (a) an over- or under-estimate? Explain using the behavior of R shown in the table, or state what additional information would be needed to decide.

(d) The reservoir holds $W(t)$ thousand gallons at time t. Water leaves the reservoir at a constant rate of 4 thousand gallons per hour during the 12-hour period. Write, but do not evaluate, an expression involving an integral for $W(12)$ given $W(0) = 80$.

FRQ 3 MEDIUM 6.4 / 6.5 Accumulation Function No Calculator

The graph of the continuous function f on $[0, 8]$ consists of three line segments and a quarter circle of radius 2, as shown. Let $g(x) = \int_0^x f(t)\,dt$.



(a) Find $g(2)$, $g(4)$, and $g(8)$.

(b) On what interval(s) is g increasing? Justify.

(c) At what value of x on $[0, 8]$ does g attain its absolute maximum? Justify.

(d) Find the x-coordinate of each point of inflection of g on $(0, 8)$. Justify.

FRQ 4 HARD 6.8 / 6.9 / 6.10 Techniques No Calculator

Evaluate each of the following integrals, showing all algebraic steps. For u-substitution problems, clearly state your choice of u and du.

(a) $\displaystyle\int \frac{2x+3}{x^2+3x+5}\,dx$

(b) $\displaystyle\int_0^1 x\sqrt{1-x^2}\,dx$

(c) $\displaystyle\int \frac{x^2+1}{x-2}\,dx$

FRQ 5 HARD 6.4 / 6.5 / 6.7 Reasoning with g No Calculator

Let f be a continuous function on $[-2, 8]$, and define $g(x) = \int_0^x f(t)\,dt$. It is known that $\int_0^3 f(t)\,dt = 6$, $\int_3^5 f(t)\,dt = -2$, and $\int_5^8 f(t)\,dt = 4$.

(a) Find $g(3)$, $g(5)$, and $g(8)$.

(b) Find $\int_{-1}^8 \left[\, 2f(t) + 1 \,\right] dt$, given that $\int_{-1}^0 f(t)\,dt = 1$.

(c) Suppose f is differentiable on $[0, 8]$, with $f(0) = 2$ and $f(8) = -3$. Let $h(x) = x \cdot f(x)$. Find $\int_0^8 \left[\, f(x) + x\, f'(x) \,\right] dx$.

(d) Let $H(x) = \int_0^{x^2} f(t)\,dt$. Express $H'(x)$ in terms of f and x.

BC-Only Practice

BC ONLY. The following items cover Integration by Parts (Topic 6.11), Linear Partial Fractions (Topic 6.12), and Improper Integrals (Topic 6.13). AB students may skip this section. BC students should be fluent with: (i) IBP $\int u\,dv = uv - \int v\,du$ — pick u via LIATE; (ii) decomposing proper rationals with distinct linear factors as $\frac{A}{x-r_1} + \frac{B}{x-r_2} + \cdots$; (iii) evaluating improper integrals as limits, including those with interior discontinuities (split first).

Q BC1 MEDIUM 6.11 IBP BC ONLY No Calculator

$$\int_0^1 x\,e^x\,dx =$$

(A) 0

(B) 1

(C) $e-1$

(D) e

Q BC2 MEDIUM 6.12 Partial Fractions BC ONLY No Calculator

The partial-fraction decomposition of $\frac{1}{x^2-1}$ is

(A) $\frac{1}{x-1} + \frac{1}{x+1}$

(B) $\frac{1/2}{x-1} - \frac{1/2}{x+1}$

(C) $\frac{1/2}{x-1} + \frac{1/2}{x+1}$

(D) $\frac{1}{x-1} - \frac{1}{x+1}$

Q BC3 EASY 6.13 Improper BC ONLY No Calculator

$$\int_1^\infty \frac{1}{x^2}\,dx =$$

(A) 0

(B) 1

(C) ∞ (diverges)

(D) $\frac{1}{2}$

Q BC4 MEDIUM 6.13 Improper (Compare) BC ONLY No Calculator

Which of the following improper integrals diverge?

I. $\displaystyle\int_1^{\infty} \frac{1}{x}\, dx$

II. $\displaystyle\int_1^{\infty} \frac{1}{x^{3/2}}\, dx$

III. $\displaystyle\int_0^{1} \frac{1}{\sqrt{x}}\, dx$

(A) I only

(B) I and II

(C) II and III

(D) I, II, and III

FRQ BC1 HARD 6.11 IBP (Twice) BC ONLY No Calculator

Evaluate each integral using integration by parts. State your choice of u and dv at every step.

(a) $\displaystyle\int x^2 e^x \, dx$ — apply IBP twice.

(b) $\displaystyle\int_1^e \ln x \, dx$ — use $u = \ln x$, $dv = dx$.

(c) Briefly explain how the LIATE mnemonic guided your u selection in parts (a) and (b).

FRQ BC2 HARD 6.12 / 6.13 Partial Fractions + Improper BC ONLY No Calculator

Consider $\displaystyle\int_2^{\infty} \frac{1}{x^2 - x}\, dx$.

(a) Decompose $\dfrac{1}{x^2 - x}$ into partial fractions. Show the algebraic system you solve.

(b) Set up the integral as a limit, evaluate the antiderivative, and determine whether the integral converges. If it converges, give the exact value.

(c) Now consider $\displaystyle\int_0^{2} \frac{1}{(x-1)^{1/3}}\, dx$, which is improper at the interior point $x = 1$. Split the integral, evaluate each piece as a one-sided limit, and find the total value (or state divergence). Justify each step.